
Michael Kummer

<https://MichaelKummer.ca> | +1 (780) 200-4824 | mkummer@ualberta.ca

EDUCATION

UNIVERSITY OF ALBERTA

SEPT 2023 – DEC 2025

Bachelor of Arts

Edmonton, Alberta

Major: Economics **Minor:** Computer Science

Focus: Data Science, Applied Microeconomic Models, Programming Methodologies, Statistical Methods.

UNIVERSITY OF ALBERTA

SEPT 2018 – MAY 2023

Bachelor of Arts

Edmonton, Alberta

Major: Psychology & History

Focus: Behaviour Analysis, Historical Analysis & Documentation, Research Methods & Statistics.

TECHNICAL SKILLS

Technical: Python (Scikit-learn, Pandas, XGBoost, FastAPI, Matplotlib, Statsmodels), Postgres, MSSQL, R, C# (xUnit, Avalonia, WinForm), Julia, Bash, Git, Excel.

Research: Statistical Analysis, Data Collection & Visualization, Data Pipeline Design, Model Evaluation, OMOP.

Tools: Docker, Jupyter Lab, WSL, Power BI, SSMS.

WORK EXPERIENCE

LORETTA HEALTH

JAN 2026– PRESENT

Data Scientist, BERLIN, GERMANY

- Built a production-ready Machine Learning pipeline for predicting diabetes risk and self-rated health.
- Automated OCR data cleaning and standardization to OMOP codes for model training compatibility.
- Sourced and consolidated 4 datasets with 500 000 observations for model cold start and trend analysis.
- Designed API endpoints to serve real-time health data to the front-end application.
- Built ensemble of 4 models with majority voting, surfacing key health drivers for personalized user recommendations.

NORTHERN GRADIENT AI

MAY 2025 – DEC 2025

Part-Time Junior Developer, EDMONTON AB

- Debugged and resolved issues in a legacy C# WinForm application on a contract/as-needed basis.
- Developed unit tests using the xUnit framework to validate code modifications and prevent regressions.
- Created a mock production environment using MSSQL, Docker, and Hyper-V for functional and UI testing.

UNIVERSITY OF ALBERTA STUDENTS UNION

SEPTEMBER 2022 - DECEMBER 2025

Admin Assistant, EDMONTON AB

- Processed up to \$75 000 in rent, event tickets, booking fees, and major event registration payments.
- Maintained weekly inventory logs for office supplies and managed student club data transfer to a new system.
- Provided frontline student support, addressing inquiries about services, events, and administrative processes.

EVENT HOROWITZ /BALMORAL LOUNGE

MAY 2022 – DECEMBER 2025

Event Staff/Bartender, EDMONTON AB

- Set up venues for 300+ guests according to client specifications and timelines.
- Delivered high-volume food and beverage service for student events and weddings.
- Worked the Balmoral Lounge, including set up, bartending, and cash out duties.

TELUS WORLD OF SCIENCE**APRIL 2024 – AUGUST 2024****Assistant Volunteer Coordinator, EDMONTON AB**

- Recruited and onboarded 50 volunteers across two recruitment cycles.
- Drafted naloxone policy to meet workplace safety requirements.
- Facilitated a seniors focus group on post-COVID changes, analyzing insights to drive targeted adaptations.
- Led volunteer research on post-COVID management to support a \$25,000 grant application.
- Assisted the People & Culture team by creating a key management system and archiving old documents.

EXPLORE EDMONTON**JUNE 2022 – AUGUST 2022****KDays Volunteer Services Summer Intern, Edmonton, AB**

- Coordinated daily schedules for 200 volunteers throughout the 10-day KDays festival.
- Collaborated with managers to optimize volunteer placement and divide roles between staff and volunteers.
- Maintained communication with the volunteer team through bi-weekly newsletters.
- Consolidated the summer's volunteer statistics using Volgistics batch reports to inform future program improvements.

UNIVERSITY OF ALBERTA STUDENTS UNION**JULY 2020 – MAY 2021****Safewalk Coordinator, Edmonton, AB**

- Managed recruitment and training of 75 volunteers using data-driven scheduling to optimize resource allocation.
- Led marketing campaigns to revamp service image, analyzing engagement metrics to boost campus visibility.
- Tracked \$10,000 budget with forecasts and variance analysis to ensure financial compliance and efficiency.
- Implemented reward programs, improving volunteer morale and retention via participation data insights.
- Developed training programs, enhancing client satisfaction through feedback collection and statistical analysis.
- Modernized volunteer system via policy overhauls supported by performance data analysis and reporting.

ESSAY PUBLICATION

KUMMER, M. THE DEMOGRAPHIC DIVIDE: AN ANALYSIS OF CANADIAN FRAUD LOSS PATTERNS, SOCIETY OF ACTUARIES ESSAY COMPETITION, 2025. (TOP 3 WINNER - \$1500 USD).

<https://www.soa.org/resources/research-reports/2025/ai-retirement-essay-collection/>

VOLUNTEER EXPERIENCE

VICTORIA SETTLEMENT**SEPTEMBER 2022 – DECEMBER 2022****Hist 460 CSL Placement Student, EDMONTON AB**

- Digitized 19th-century land titles and organized homestead addresses into Excel.
- Synthesised land title research into a presentation for use in tours.

RELEVANT PROJECTS (Available on MichaelKummer.ca)

VOLUNTEER CONNECTOR PROJECT**2025**

- Automated daily capture of volunteer postings across Canada via Volunteer Connector API using the requests library.
- Built SQLite3 database with automated data pipeline using cron jobs for daily capture.
- Developed interactive Plotly dashboard to visualize category trends and analyze posting patterns.

REENGAGING SENIOR VOLUNTEERS POST-COVID**2025**

- Analyzed age-related barriers to volunteerism using Alberta government survey data (n=1,564).
- Performed statistical testing to identify significant differences in volunteer challenges between age cohorts.
- Built interactive data visualizations and conducted demographic segmentation analysis using Python statistical libraries.
- Translated research findings into targeted strategies for addressing post-COVID volunteer engagement challenges.